

Assignment 03 Final Report - NYC Taxi ETL & Analytics

This report summarizes ETL traceability from A2, row-count validation, analytical queries, business insights, visualizations, and optimization notes for the warehouse pipeline.

A2 Traceability Table

A2 Quality Issue	Transformation Applied	PySpark Function(s)
Duplicate rows	dropDuplicates()	dropDuplicates
Negative fare_amount	Distance-bin median imputation	when + percentile_approx + join
Zero/non-integer passenger_count	round and floor at min=1	round + greatest
Trip distance outliers	Cap at 99th percentile	percentile_approx + when
Tip greater than fare	Cap tip at cleaned fare	when
Datetime standardization	to_timestamp conversions	to_timestamp
Missing numeric values	Median imputation	coalesce + percentile_approx

Row Count Verification

Raw input rows: 2,964,624
fact_trips rows: 2,964,624
dim_datetime rows: 749
dim_payment rows: 5
dim_trip_category rows: 3

No unexplained row-loss was observed at fact level in this run. Dimension row-count differences are expected due to deduplication and dimensional modeling.

Q1 Monthly revenue trend

year_month	total_revenue	total_trips
2002-12	13.0	2
2009-01	100.0	3
2023-12	170.37	10
2024-01	65094003.18000247	2964606
2024-02	64.28	3

Q1 Monthly revenue trend - Business Interpretation

Monthly revenue and trip totals quantify the overall demand cycle and provide a baseline for capacity planning. A month with higher revenue but relatively flat trip volume usually signals a higher average fare per trip, often due to longer distances or pricing dynamics. The business can use this trend to plan budgets, allocate drivers, and schedule maintenance in lower-revenue windows. This view also supports KPI tracking for revenue per trip and demand growth over time.

Q2 Top hours by revenue (RANK)

pickup_hour	hour_revenue	revenue_rank
17	4484362.179999994	1
16	4421529.560000003	2
18	4361090.240000009	3
15	4271110.030000005	4
14	4157907.5300000096	5
19	3906230.1300000176	6
13	3694523.9500000086	7
21	3533936.3099999945	8
20	3469064.0799999903	9
12	3455546.060000005	10
22	3263941.5499999924	11
11	3143167.2000000146	12
10	2953171.2700000065	13
9	2731800.910000009	14
23	2618241.799999996	15
8	2460905.7300000163	16
7	1837409.470000008	17
0	1836556.870000003	18
1	1128778.5600000012	19
6	1046710.1200000048	20

Q2 Top hours by revenue (RANK) - Business Interpretation

Hour-level ranking identifies the most valuable operating windows where both customer demand and fare generation are strongest. These ranked periods are ideal for surge-pricing policy reviews and targeted driver incentives. Operations teams can align shift start times to these peak hours to reduce idle time and increase trip throughput. Marketing can also schedule promotional offers in low-ranked hours to smooth demand.

Q3 Payment behavior trend (LAG)

payment_desc	pickup_year	pickup_month	trips	prev_month_trips
Cash	2009	1	3	nan
Cash	2023	12	2	3.0
Cash	2024	1	439185	2.0
Cash	2024	2	1	439185.0
Credit card	2023	12	8	nan
Credit card	2024	1	2319036	8.0
Credit card	2024	2	2	2319036.0
Dispute	2024	1	46628	nan
No charge	2002	12	2	nan
No charge	2024	1	19595	2.0
Other	2024	1	140162	nan

Q3 Payment behavior trend (LAG) - Business Interpretation

The lag-based month-over-month comparison reveals whether each payment channel is growing, stable, or declining. A consistent increase in card usage supports investment in digital checkout reliability and fraud controls. If cash usage grows in specific periods, finance teams can adjust reconciliation and risk workflows accordingly. This trend also helps prioritize customer experience improvements by payment segment.

Q4 Distance bucket profitability

distance_bucket	trip_count	avg_revenue_per_trip	avg_duration_min
long	350473	64.21	36.99
medium	890011	23.49	19.32
short	1724140	12.58	9.35

Q4 Distance bucket profitability - Business Interpretation

Distance buckets show profitability differences across short, medium, and long trips. Comparing average revenue with average duration highlights which trip type generates better return per unit time. This can inform dispatch policy, for example favoring high-yield segments in constrained periods. It also supports pricing calibration and targeted service-level strategies for different route lengths.

Q5 Vendor-hour productivity (ROW_NUMBER)

VendorID	pickup_hour	revenue	trips	rn
1	17	1015105.9700000028	50068	1
1	16	1012609.0300000024	46988	2
1	15	1007554.3800000044	47854	3
1	14	1005689.8000000004	46892	4
1	18	963295.6400000032	50114	5
1	13	911100.2800000028	44025	6
1	12	859257.4600000039	42395	7
1	19	815509.3000000033	41863	8
1	11	799921.1200000023	39037	9
1	10	747145.3500000027	36698	10
1	9	715079.8000000014	35062	11
1	20	706658.2400000006	35737	12
1	21	705273.1000000006	35592	13
1	8	668171.2300000009	33045	14
1	22	634999.0	30733	15
1	7	525000.55	25237	16
1	23	503021.34000000014	22958	17
1	0	353274.18000000017	16692	18
1	6	325935.7699999998	13605	19
1	1	228914.30000000025	11477	20

Q5 Vendor-hour productivity (ROW_NUMBER) - Business Interpretation

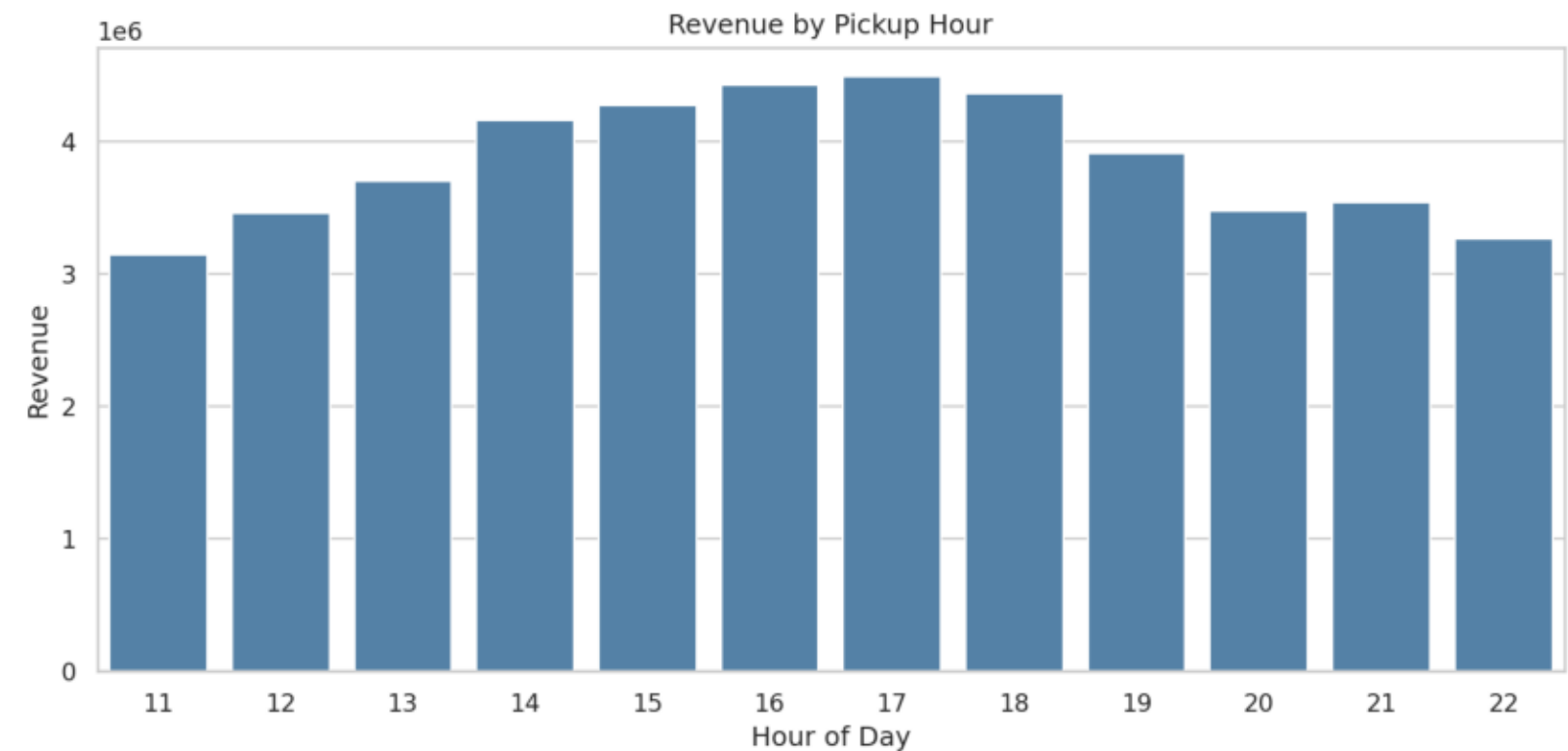
Vendor-hour productivity ranking surfaces the strongest revenue combinations for each vendor. Top-ranked hours represent opportunities to concentrate resources where vendor performance is highest. Low-ranked windows can be examined for route mix, pickup density, or operational inefficiencies. This evidence supports vendor benchmarking and incentive design based on measurable productivity.

Line Chart - Revenue Trend



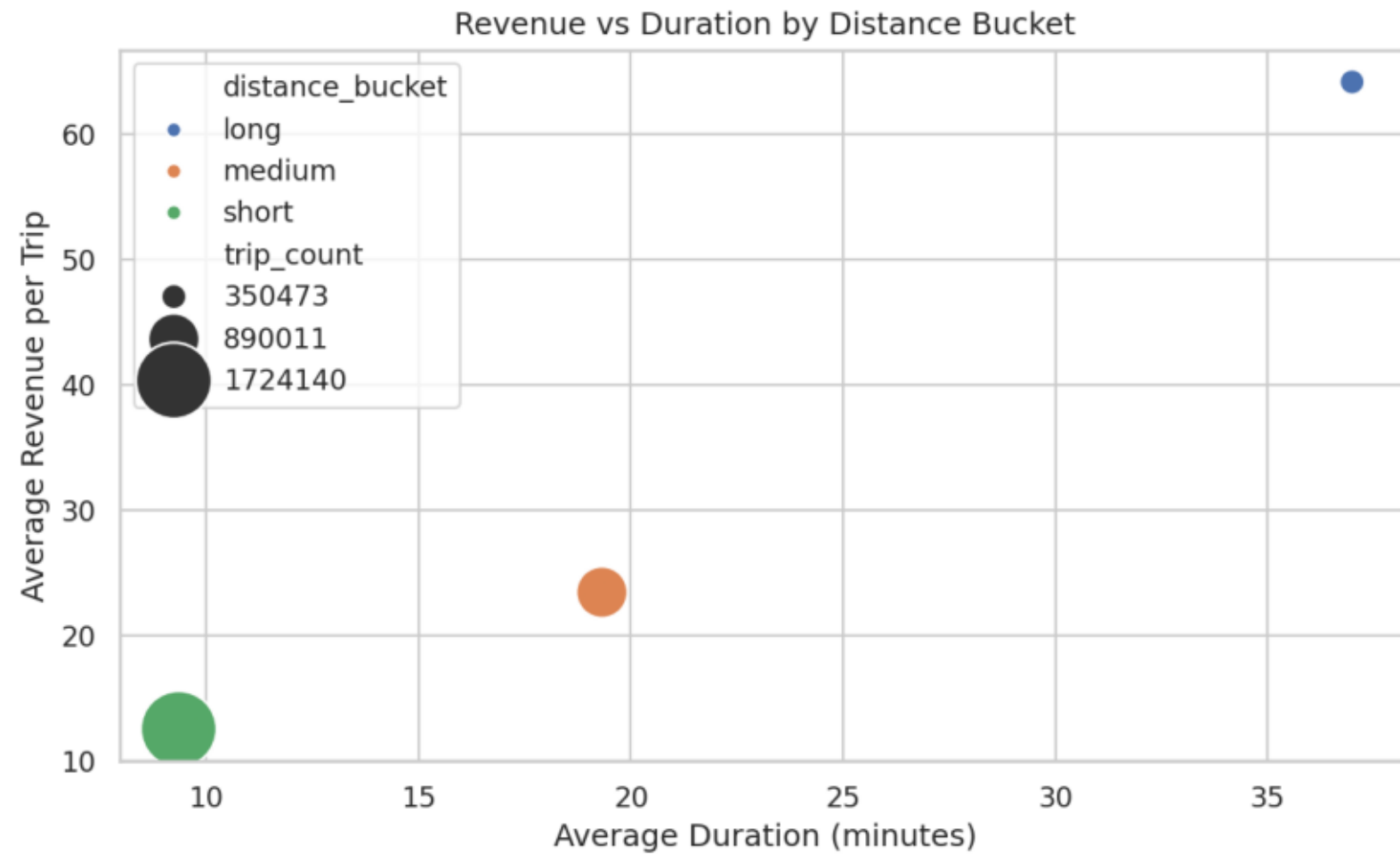
The trend line shows how total revenue evolves over time. It helps identify growth or contraction periods that should drive planning decisions. Teams can use this view for monthly forecasting and performance tracking.

Bar Chart - Hourly Revenue



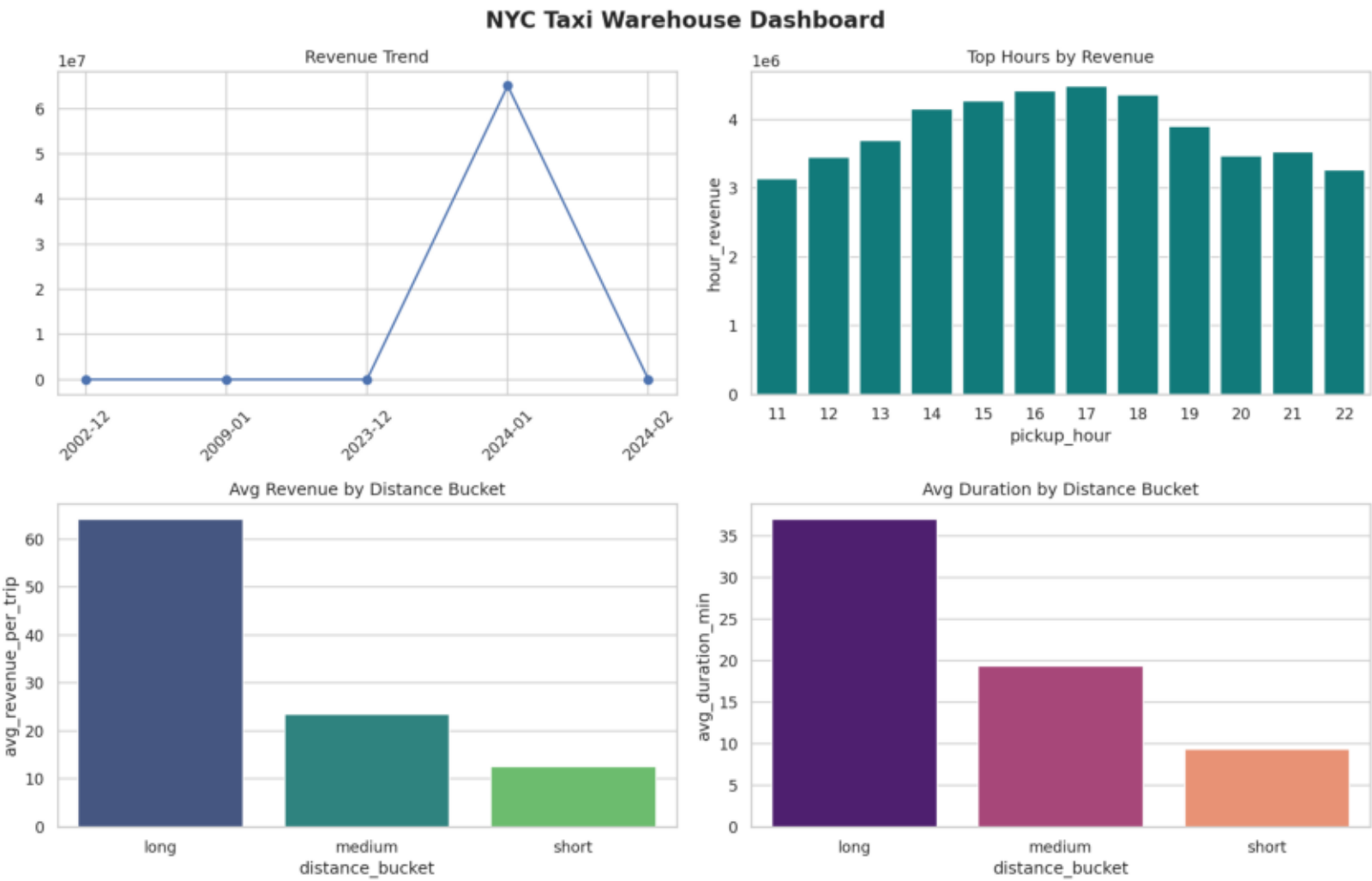
The hourly bars compare revenue concentration across the day. Higher bars identify priority windows for fleet availability and staffing. Lower bars indicate where promotions or routing improvements may be needed.

Scatter Chart - Revenue vs Duration



This scatter view compares average trip duration with average revenue by distance class. It highlights which segments are efficient versus time-consuming for the revenue produced. The business can use this to tune dispatch and pricing strategies by trip type.

Dashboard (3+ subplots)



The dashboard combines trend, hourly, and bucket-level perspectives in one place. Cross-reading these subplots helps identify whether revenue shifts come from timing, trip mix, or both. This integrated view supports fast operational decision-making.

Pipeline Optimization Summary

Optimization techniques used:

- 1) Partitioning: fact and datetime tables are partitioned by pickup_year/pickup_month.
- 2) Caching: transformed DataFrame is cached before repeated dimension/fact derivations.
- 3) Broadcast joins: small dimensions are broadcast-joined to fact to reduce shuffle.
- 4) Query plan analysis: use Spark .explain(True) on complex window query during review.

HDFS Processed Directory Screenshot

```
C:\Windows\System32>hdfs dfs -ls -R /warehouse/raw/nyc_taxi/year=2024/month=01  
-rw-r--r--    1 hassa supergroup  49961641 2026-04-19 20:09 01/yellow_tripdata_2024-01.parquet  
  
C:\Windows\System32>
```

Screenshot evidence of warehouse directory structure and data files in HDFS.